PSGR
Krishnammal College for Women

Affiliated to Bharathiar University \ Autonomous \ College of Excellence \ Accredited with A++ Grade \ Ranked 9th in NIRF 2025

College of Excellence, NIRF 2025- 9thRank

Autonomous and Affiliated to Bharathiar University

Accredited with A++ grade by NAAC, An ISO 9001:2015 Certified Institution

Peelamedu, Coimbatore – 641004.

INSTAGRAM INFLUENCER PERFORMANCE ANALYSIS

Case Study work submitted to PSGR Krishnammal College for women
in partial fulfilment of the requirements for the award of the degree of
Master of Science in Data Analytics
Bharathiar University, Coimbatore – 641046

Done By

SUBHIKSHA N

25SMDA049

Guided By

Dr.K.GANDHIMATHI M.Sc., M.Phil., Ph.D.

ASSISTANT PROFESSOR

DEPARTMENT OF DATA ANALYTICS (PG)

PSGR KRISHNAMMAL COLLEGE FOR WOMEN

Coimbatore-641004.

www.psgrkcw.ac.in

AUGUST 2026

CONTENT

S NO	TITLE	PG NO
	ABSTRACT	**I**
1	**INTRODUCTION**	**1**
	1.1 Literature Review	**2**
	1.2 Problem Statement	**3**
	1.3 Tool Description	**4**
2	**DOMAIN- INSTAGRAM INFLUENCER ANALYTICS**	**7**
3	**METHODOLOGY**	**9**
	3.1 Process Flow	**9**
	3.2 Data collection	**10**
	4.3 Dataset Description	**11**
	4.4 Data Transformation	**15**
	4.5 DAX Measures	**16**
4	**INFLUENCER PERFORMANCE ANALYSIS-DASHBOARD**	**18**
	4.1 Dashboard Design	**18**
	5.2 Dashboard Creation	**19**
5	**RESULTS & DISCUSSION**	**25**
6	**CONCLUSION**	**26**
7	**REFERENCES**	**27**

ABSTRACT

Social media platforms generate vast amounts of data related to influencer performance, audience engagement, content preferences, and demographic reach. However, extracting meaningful insights from this data can be challenging due to the large number of metrics and categories involved. This project addresses this challenge by developing an Instagram Influencer Categorical Performance Analysis Dashboard in Microsoft Power BI, demonstrating how influencer data can be transformed into an interactive Business Intelligence solution.

Dataset containing 20,000 influencer records was utilized, incorporating information on creator profiles, follower counts, engagement rates, authentic engagement, audience demographics, content formats, creator tiers, verification status, audience quality, and geographic reach. Power Query was used for data preparation and transformation, while DAX measures were developed to calculate key performance indicators and analytical metrics. These measures provide a consistent foundation for all visualizations and performance analyses presented within the dashboard.

The completed solution consists of an Executive Performance Analysis page and ten category-specific analytical pages covering Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Tech, and Travel influencers. Through KPI cards, interactive visualizations, demographic analysis, audience insights, creator tier evaluation, and comparative performance analysis, the dashboard enables users to identify trends, evaluate influencer effectiveness, and compare performance across categories. The resulting solution supports data-driven decision-making and demonstrates the application of Business Intelligence techniques in influencer marketing analytics.

1.INTRODUCTION

Social media platforms generate enormous volumes of data through influencer profiles, content creation activities, audience interactions, likes, comments, followers, and brand collaborations. The rapid expansion of influencer marketing has transformed the way businesses promote products, strengthen brand visibility, and engage with target audiences. As organizations increasingly rely on influencers for digital marketing campaigns, the ability to analyze influencer performance accurately has become essential for effective decision-making. However, influencer-related information is often distributed across spreadsheets, social media analytics tools, and multiple reporting sources, making it difficult to obtain a unified view of performance. Traditional evaluation methods that focus primarily on follower counts are frequently inadequate because they do not fully reflect audience engagement, content effectiveness, authenticity, or overall influence. Manual reporting processes are also time-consuming and often fail to provide the dynamic insights required for influencer selection and campaign optimization.

The analysis is performed using social_media_Instagram_Influencers_final.xlsx dataset, which contains 20,000 Instagram influencer records represented through 17 attributes covering profile information, audience demographics, and performance indicators. Key attributes include influencer identity, influencer category, audience country, gender, age group, creator tier, followers, average likes, average comments, engagement rate, authentic engagement, content format, verification status, and audience quality. These attributes provide a comprehensive representation of influencer reach, audience behavior, and content performance. The dataset supports comparison across ten major influencer categories—Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel—enabling a detailed analysis of performance variations across different influencer segments and audience groups.

Data preparation was performed using Power Query Editor within Microsoft Power BI Desktop. This process involved validating data quality, identifying duplicate records, checking for missing values, assigning appropriate data types, and standardizing numerical attributes. Performance-related metrics such as followers, likes, comments, engagement rate, and authentic engagement were reviewed to ensure consistency and accuracy. Additional calculated fields were created where necessary to support ranking, comparison, and categorical analysis. These

preprocessing activities established a reliable and structured dataset capable of supporting advanced analytical reporting.

To enhance analytical functionality, the dashboard incorporates DAX (Data Analysis Expressions) measures to calculate important Key Performance Indicators (KPIs), including Total Influencers, Total Followers, Average Engagement Rate, Total Authentic Engagement, Average Likes, Average Comments, Highest Performing Influencer, Lowest Performing Influencer, Audience Distribution, Gender Performance, Age Group Performance, and Creator Tier Analysis. These centralized calculations ensure consistency across report pages while improving scalability, maintainability, and real-time reporting based on user-selected filters.

The completed dashboard consists of multiple interactive report pages designed to support both overview and detailed analysis. The Overview Dashboard presents high-level insights through KPI cards, engagement metrics, audience distribution, and category comparisons. Additional dashboards provide detailed evaluation of influencer reach, audience quality, engagement performance, creator tiers, demographic characteristics, and content effectiveness within each Influencer. Visualizations such as clustered bar charts, column charts, line charts, donut charts, treemaps, scatter plots, maps, and slicers enable flexible exploration of the data. Overall, the Instagram Influencer Categorical Performance Analysis Dashboard demonstrates how Business Intelligence techniques transform large-scale social media data into actionable insights, helping organizations identify high-performing influencer categories, understand audience engagement behavior, optimize influencer selection, and support strategic marketing decisions through accurate, data-driven analysis.

1.2 Literature review

Freberg, K. et.al [5] (2011), examined the role of social media influencers and their impact on communication between brands and audiences. The study highlighted that influencers act as important digital opinion leaders, and their effectiveness depends on factors such as credibility, audience relationship, and engagement levels.

Yuan, S. [6] (2019), investigated the influence of social media influencers on consumer trust and branded content effectiveness. The findings showed that influencer credibility, content

value, and audience engagement significantly affect how users perceive promotional content and interact with brands.

De Veirman, M. et.al [7] (2017), analyzed the relationship between influencer characteristics and brand engagement on Instagram. The study found that influencer popularity, follower count, and content presentation influence audience responses, but higher follower numbers alone do not always guarantee better engagement performance.

Eckerson, W. [8] (2011), explained the principles of effective performance dashboard design and data visualization techniques. The study highlighted that interactive dashboards improve decision-making by presenting important Key Performance Indicators (KPIs) through clear, visual, and user-friendly analytical interfaces.

Russo, M. et.al [9] (2015), provided detailed insights into Data Analysis Expressions (DAX) used in Microsoft Power BI, SQL Server Analysis Services, and Excel. The authors emphasized that well-designed DAX measures improve calculation accuracy, maintain consistency, and support dynamic analysis across multiple dashboard reports.

1.2 Problem Statement

The management and evaluation of Instagram influencer performance data in digital marketing environments is characterized by fragmented data sources that limit the speed, accuracy, and reliability of influencer-based decision-making. Influencer profile details, audience information, content categories, follower statistics, engagement metrics, and performance indicators are typically maintained across separate datasets or manually prepared reports, each collected and updated independently. When businesses need a consolidated answer to a marketing-related question — for example, which influencer category generates the highest engagement, whether audience quality contributes to better performance, or which creator tier provides stronger interaction levels — the process of collecting figures from different sources and comparing their formats, metrics, and performance parameters becomes time-consuming and may produce inconsistent results.

Such fragmentation produces several specific and recurring problems. First, no single, standardized definition exists for measuring influencer success, as different organizations may

evaluate creators using different metrics such as follower count, likes, comments, engagement rate, or authentic engagement. These different evaluation approaches can produce conflicting results and make it difficult to accurately identify high-performing influencers. Second, without a unified analytical platform, businesses have limited ability to perform cross-dimensional analysis, such as examining whether specific Influencers generate higher engagement rates, whether certain audience groups interact more effectively with particular content categories, or whether smaller creator tiers achieve better audience engagement compared to large-scale influencers. These insights are essential for developing targeted influencer marketing strategies.

Existing influencer reporting methods are often manually prepared and therefore produced infrequently, making the analysis outdated when compared with rapidly changing social media trends. In an environment where audience behavior, content preferences, and engagement patterns continuously change, static reports provide limited value for the daily and strategic decisions required by marketers. Fourth, the absence of interactive filtering in spreadsheet-based analysis means that exploring a new question, such as the performance of a specific Influencer, audience country, gender group, or creator tier, requires additional manual calculations rather than a simple filter selection within an existing analytical system. The developed dashboard directly addresses these challenges by consolidating influencer profile data, audience characteristics, and performance metrics into a single governed Power BI model. Every metric is defined using centralized DAX calculations, every dashboard page is interactively filterable, and every visualization updates dynamically based on user selections, enabling efficient and reliable influencer performance analysis.

1.3 Tool Description

The dashboard was developed using Microsoft Power BI Desktop, with Microsoft Excel serving as the source data application. The project was created and executed in a 64-bit Windows environment using the required hardware and software specifications. The following table summarizes the hardware and software environment used to develop and run the Instagram Influencer Categorical Performance Analysis Dashboard.

Table 1.3.1: Software and Hardware Requirements

Category	Specification
Processor	13th Gen Intel(R) Core(TM) i5-1335U (1.30 GHz)
RAM	16 GB (15.7 GB usable)
System Type	64-bit Operating System, x64-based processor
Operating System	Windows 10/11 64-bit
BI Authoring Tool	Microsoft Power BI Desktop
Data Source Application	Microsoft Excel (social_media_Instagram_final.xlsx)
Calculation Language	DAX (Data Analysis Expressions)
Data Transformation Tool	Power Query Editor (M language) within Power BI Desktop

Table 1.3.1 presents the software and hardware requirements necessary for developing and running the Instagram Influencer Categorical Performance Analysis Dashboard using Power BI

Microsoft Power BI

Microsoft Power BI is a Business Intelligence and data visualization platform used to transform raw data into interactive dashboards and analytical reports. Power BI Desktop was used to import the Instagram influencer dataset, perform data transformation, create DAX measures, and develop interactive dashboards for Analyzing influencer performance, engagement patterns, audience behavior, and category-wise comparisons.

Microsoft Excel

Microsoft Excel was used as the source data application for the dashboard. The workbook social_media_Instagram_Influencers_final.xlsx contains influencer details, audience characteristics, engagement metrics, content formats, and performance attributes. The dataset was imported into Power BI, where it was prepared and connected for analytical reporting.

Power Query Editor

Power Query Editor is the data transformation tool available within Power BI Desktop. It was used to import the Excel dataset, remove unnecessary fields, handle missing values, check data consistency, and assign appropriate data types. These transformations prepared the dataset for accurate analysis and dashboard development.

Data Analysis Expressions (DAX)

Data Analysis Expressions (DAX) is the formula language used in Power BI to create measures, calculated values, and KPIs. DAX was used to calculate influencer performance metrics such as total influencers, followers, engagement rate, authentic engagement, and category-based analysis measures, ensuring consistent calculations throughout the dashboard.

2. DOMAIN – INSTAGRAM INFLUENCER ANALYTICS

The domain of this report is Instagram Influencer Analytics, with a primary focus on Social Media Business Intelligence and digital marketing performance measurement. Instagram influencer analytics enables businesses and marketers to convert regularly collected social media data - influencer profiles, follower information, audience characteristics, content categories, engagement metrics, and interaction behavior - into meaningful insights that support influencer selection, campaign planning, audience targeting, and marketing performance evaluation. By Analyzing this data systematically rather than depending on basic follower counts or manual comparisons, organizations can identify effective influencer strategies and make data-driven decisions for digital marketing activities.

Instagram influencer analytics plays a significant role in evaluating creator performance by examining metrics such as follower reach, engagement rate, authentic engagement, average likes, average comments, audience demographics, Influencer category performance, creator tier, and content effectiveness. Analyzing these indicators together rather than individually enables businesses to understand which influencer categories generate higher audience interaction, which creator segments provide better engagement, and how audience characteristics influence influencer success. This approach helps organizations identify high-performing creators and develop more effective influencer marketing strategies.

The work also belongs to the broader field of Business Intelligence, where Microsoft Power BI transforms raw influencer datasets into interactive dashboards and analytical reports. Business Intelligence enables marketers and analysts to monitor Key Performance Indicators, compare Influencer-wise performance, evaluate engagement behavior, analyze audience distribution, and identify performance trends through dynamic reports. Instead of relying on static spreadsheets or manually prepared summaries, Power BI provides an interactive environment where insights can be explored through filters, slicers, and visual analysis based on changing user requirements.

A further component of this domain is Data Visualization, which represents complex influencer performance data through charts, KPI cards, maps, treemaps, scatter plots, and interactive dashboards. Effective visualization allows marketers to understand patterns such as which Influencers achieve higher engagement rates, how followers are distributed across countries, how creator tiers perform, and how audience demographics influence engagement behavior

without directly Analyzing large datasets. Selecting appropriate visualizations based on the analytical objective — such as comparison, distribution, trend analysis, correlation, or ranking forms an important part of influencer dashboard design.

Digital Marketing Analytics focuses on measuring and optimizing marketing performance through social media data. While influencer analytics identifies creator performance and audience response, digital marketing analytics emphasizes the application of these insights to improve promotional strategies and campaign outcomes. Category-based dashboards for Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel influencers demonstrate the integration of these domains by transforming influencer data into actionable marketing insights.

Taken together, these related domains — Instagram Influencer Analytics, Social Media Business Intelligence, Data Visualization, and Digital Marketing Analyti**cs** — provide the conceptual foundation for the data model, DAX measure layer, and interactive Power BI dashboards developed throughout this report. These domains support the systematic evaluation of influencer performance and demonstrate how Business Intelligence techniques can be applied to improve decision-making in the digital marketing environment.

3.METHODOLOGY

The methodology involves collecting and preparing the Instagram influencer dataset by cleaning, transforming, and validating the data for analysis. The processed data is imported into Microsoft Power BI, where data modeling and DAX measures are created to calculate key performance metrics. Interactive dashboards are designed using charts, KPI cards, slicers, and filters to analyze influencer performance across different categories. The dashboards provide meaningful insights into audience engagement, demographics, and influencer effectiveness, supporting data-driven marketing decisions.

3.1Process Flow

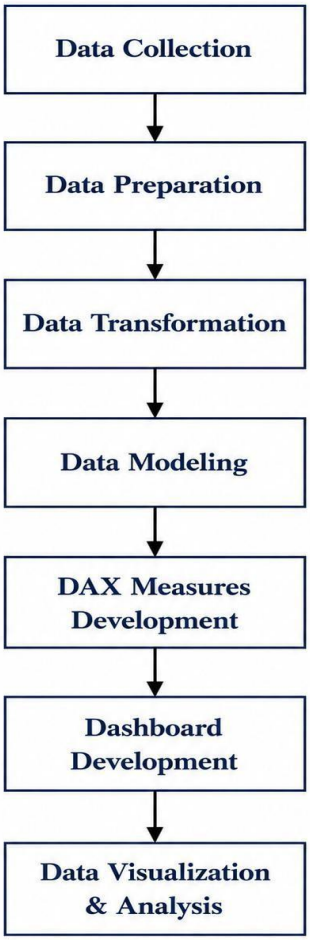


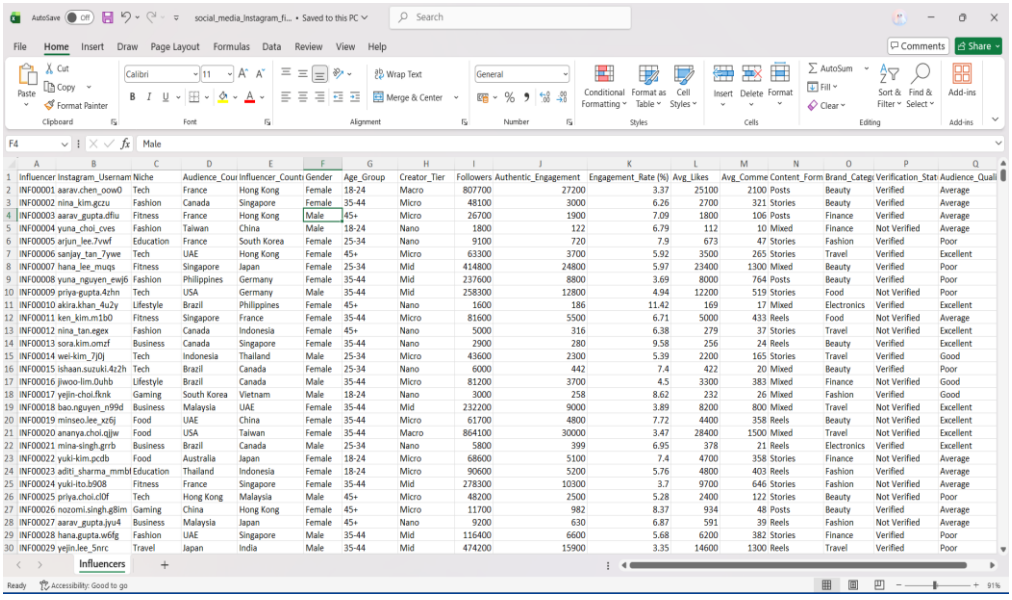
Fig 3.1.1: Instagram Influencer Analytics Dashboard Development Process Flow

Figure 3.1.1 illustrates the workflow followed for developing the Instagram Influencer Categorical Performance Analysis Dashboard in Microsoft Power BI. The process includes data preparation, transformation, data modelling, DAX measure creation, dashboard development, and visualization to analyze influencer performance and engagement patterns.

3.2 Data collection

The data used was collected from *social_media_Instagram_Influencers_final.xlsx*, a structured Microsoft Excel workbook containing 20,000 Instagram influencer records with multiple attributes related to influencer profiles, audience characteristics, content categories, and performance metrics. The dataset represents influencer marketing information, covering creator details, Instagram usernames, influencer categories, audience demographics, follower statistics, engagement behavior, content formats, verification status, and audience quality indicators.

The dataset was imported into Microsoft Power BI Desktop as a structured table using Power BI's native Excel connector. The imported dataset preserves the original attribute-level structure of the source workbook, allowing the creation of a semantic model suitable for influencer performance analysis rather than converting the information into an unstructured format**.**



Influencer	Instagram_Usernam	Niche	Audience_Cou	Influencer_Counti	Gender	Age_Group	Creator_Tier	Followers	Authentic_Engagement	Engagement_Rate (%)	Avg_Likes	Avg_Comme	Content_Form	Brand_Categ	Verification_Stat	Audience_Quali
INF00001	aarav.chen_oow0	Tech	France	Hong Kong	Female	18-24	Macro	807700	27200	3.37	25100	2100	Posts	Beauty	Verified	Average
INF00002	nina_kim.gczu	Fashion	Canada	Singapore	Female	35-44	Micro	48100	3000	6.26	2700	321	Stories	Beauty	Verified	Average
INF00003	aarav_gupta.dfiu	Fitness	France	Hong Kong	Male	45+	Micro	26700	1900	7.09	1800	106	Posts	Finance	Verified	Average
INF00004	yuna_choi_cves	Fashion	Taiwan	China	Male	18-24	Nano	1800	122	6.79	112	10	Mixed	Finance	Not Verified	Average
INF00005	arjun_lee.7vwf	Education	France	South Korea	Female	25-34	Nano	9100	720	7.9	673	47	Stories	Fashion	Verified	Poor
INF00006	sanjay_tan_7ywe	Tech	UAE	Hong Kong	Female	45+	Micro	63300	3700	5.92	3500	265	Stories	Travel	Verified	Excellent
INF00007	hana_lee_muqs	Fitness	Singapore	Japan	Female	25-34	Mid	414800	24800	5.97	23400	1300	Mixed	Beauty	Verified	Poor
INF00008	yuna_nguyen_ewj6	Fashion	Philippines	Germany	Female	35-44	Mid	237600	8800	3.69	8000	764	Posts	Beauty	Verified	Poor
INF00009	priya-gupta.4zhn	Tech	USA	Germany	Male	35-44	Mid	258300	12800	4.94	12200	519	Stories	Food	Not Verified	Poor
INF00010	akira.khan_4u2y	Lifestyle	Brazil	Philippines	Female	45+	Nano	1600	186	11.42	169	17	Mixed	Electronics	Verified	Excellent
INF00011	ken_kim.m1b0	Fitness	Singapore	France	Female	35-44	Micro	81600	5500	6.71	5000	433	Reels	Food	Not Verified	Average
INF00012	nina_tan.egex	Fashion	Canada	Indonesia	Female	45+	Nano	5000	316	6.38	279	37	Stories	Travel	Not Verified	Excellent
INF00013	sora.kim.omzf	Business	Canada	Singapore	Female	35-44	Nano	2900	280	9.58	256	24	Reels	Beauty	Verified	Excellent
INF00014	wei-kim_7j0j	Tech	Indonesia	Thailand	Male	25-34	Micro	43600	2300	5.39	2200	165	Stories	Travel	Verified	Good
INF00015	ishaan.suzuki.4z2h	Tech	Brazil	Canada	Female	25-34	Nano	6000	442	7.4	422	20	Mixed	Beauty	Verified	Poor
INF00016	jiwoo-lim.0uhb	Lifestyle	Brazil	Canada	Male	35-44	Micro	81200	3700	4.5	3300	383	Mixed	Finance	Not Verified	Good
INF00017	yejin-choi.fknk	Gaming	South Korea	Vietnam	Male	18-24	Nano	3000	258	8.62	232	26	Mixed	Fashion	Verified	Good
INF00018	bao.nguyen_n99d	Business	Malaysia	UAE	Female	35-44	Mid	232200	9000	3.89	8200	800	Mixed	Travel	Not Verified	Excellent
INF00019	minseo.lee_xz6j	Food	UAE	China	Female	35-44	Micro	61700	4800	7.72	4400	358	Reels	Beauty	Not Verified	Excellent
INF00020	ananya.choi.qjjw	Food	USA	Taiwan	Female	35-44	Macro	864100	30000	3.47	28400	1500	Mixed	Travel	Not Verified	Excellent
INF00021	mina-singh.grrb	Business	Brazil	Canada	Male	25-34	Nano	5800	399	6.95	378	21	Reels	Electronics	Verified	Excellent
INF00022	yuki-kim.gcdb	Food	Australia	Japan	Female	18-24	Micro	68600	5100	7.4	4700	358	Stories	Finance	Not Verified	Average
INF00023	aditi_sharma_mmbl	Education	Thailand	Indonesia	Female	18-24	Micro	90600	5200	5.76	4800	403	Reels	Fashion	Verified	Average
INF00024	yuki-ito.b908	Fitness	France	Singapore	Female	35-44	Mid	278300	10300	3.7	9700	646	Stories	Fashion	Not Verified	Average
INF00025	priya.choi.cl0f	Tech	Hong Kong	Malaysia	Male	45+	Micro	48200	2500	5.28	2400	122	Stories	Beauty	Not Verified	Poor
INF00026	nozomi.singh.g8im	Gaming	China	Hong Kong	Female	45+	Micro	11700	982	8.37	934	48	Posts	Beauty	Verified	Average
INF00027	aarav_gupta.jyu4	Business	Malaysia	Japan	Female	45+	Nano	9200	630	6.87	591	39	Reels	Fashion	Not Verified	Average
INF00028	hana.gupta.w6fg	Fashion	UAE	Singapore	Male	35-44	Mid	116400	6600	5.68	6200	382	Stories	Finance	Verified	Poor
INF00029	yejin.lee_5nrc	Travel	Japan	India	Male	35-44	Mid	474200	15900	3.35	14600	1300	Reels	Travel	Verified	Poor

Fig 3.2.1: Sample Instagram Influencer Dataset

Figure 3.2.1 illustrates the sample Instagram influencer dataset used for developing the Power BI dashboard. It contains influencer-related information such as influencer name, Instagram username, Influencer category, audience country, followers, engagement rate, average likes, average comments, authentic engagement, and other performance-related attributes, serving as the primary data source for analysis and visualization.

3.3 Dataset Description

Table 3.3.1: Influencer Profile and Audience Table

Column	Data Type	Description
influencer_name	Text	Name of the influencer associated with the record.
instagram_username	Text	Unique Instagram username used for creator identification.
Influencer_category	Text	Influencer category such as Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel.
audience_country	Text	Primary country distribution of the influencer audience.
gender	Text	Gender classification associated with influencer or audience analysis.
age_group	Text	Audience age category used for demographic analysis.
creator_tier	Text	Influencer classification based on follower range and creator level.
verification_status	Text	Indicates whether the influencer account is verified.
audience_quality	Text	Represents the quality level of influencer audience engagement.

Table 3.3.1 presents the data dictionary for the Influencer Profile and Audience table used in the influencer analytics dashboard. It describes each column, its data type, and its purpose, providing a clear understanding of influencer identity, category information, and audience characteristics used for analysis and visualization in Power BI.

Table 3.3.2: Engagement and Performance Table

Column	Data Type	Description
followers	Whole Number	Total number of followers associated with the influencer profile.
average_likes	Whole Number	Average number of likes received per content post.
average_comments	Whole Number	Average number of comments received per content post.
engagement_rate	Decimal Number	Percentage-based measurement of audience interaction with influencer content.
authentic_engagement	Whole Number	Estimated genuine audience interaction generated by influencer content.
content_format	Text	Type of content format used by influencers.
performance_category	Text	Classification used for comparing influencer performance levels.

Table 3.3.2 presents the data dictionary for the Engagement and Performance table used in the influencer dashboard. It explains the columns, data types, and their descriptions, providing an understanding of engagement behavior, content effectiveness, and performance indicators used for analytical reporting in Power BI.

Table 3.3.3: Category and Analytical Attributes Table

Column	Data Type	Description
influencer_category	Text	Main influencer category used for categorical performance comparison.
audience_country	Text	Geographic distribution used for audience analysis and mapping.
gender	Text	Demographic attribute used for gender-based performance analysis.
age_group	Text	Audience segmentation attribute used for engagement comparison.
creator_tier	Text	Creator classification used to compare influencer levels.
verification_status	Text	Account verification indicator used for performance comparison.
audience_quality	Text	Quality indicator used to evaluate audience authenticity.

Table 3.3.3 presents the data dictionary for the Category and Analytical Attributes table used in the influencer analytics dashboard. It describes the attributes, data types, and their purposes, providing a clear understanding of category-based analysis, audience segmentation, and influencer comparison performed in Power BI.

Table 3.3.4: Calculated Metrics and Dashboard Measures

Measure	Description
Total Influencers	Calculates the total number of influencer records available for analysis.
Total Followers	Calculates the combined follower reach of all influencers.
Average Engagement Rate	Measures the average interaction percentage generated by influencer content.
Total Authentic Engagement	Calculates the total genuine engagement generated across influencer profiles.
Average Likes	Calculates the average likes received by influencer content.
Average Comments	Calculates the average comments received by influencer content.
Highest Performing Influencer	Identifies the category generating the strongest performance.
Category-wise Engagement	Compares engagement performance across different influencer Influencers.

Table 3.3.4 presents the major calculated measures developed using DAX in Microsoft Power BI for the Instagram Influencer Performance Analysis Dashboard. These measures form the foundation of the dashboard by supporting KPI calculations, category-wise performance evaluation, and interactive analysis of influencer data. They enable the computation of key metrics such as total influencers, total followers, average engagement rate, average likes, average comments, and authentic engagement.

3.4 Data Transformation

The Instagram Influencers dataset was imported into Microsoft Power BI Desktop using the Excel connector and processed through the Power Query Editor to prepare the data for analysis. During the data transformation process, column headers were promoted, unnecessary columns were removed, duplicate records were identified and eliminated, and missing values were checked to ensure data completeness and consistency. Data quality validation was performed to minimize errors and improve the reliability of the analytical results.

Appropriate data types were assigned to each attribute based on its purpose. Categorical fields such as Influencer Category, Creator Tier, Content Format, Verification Status, and Audience Quality were maintained as text, while numerical attributes including Followers, Average Likes, Average Comments, Authentic Engagement, and Engagement Rate were converted into suitable numeric and percentage formats to support accurate calculations and visualizations. In addition, abbreviated values represented in thousands (K) and millions (M) were transformed into their corresponding numeric values, enabling proper aggregation and comparison across influencers.

To maintain consistency throughout the dataset, influencer categories were standardized into predefined groups such as Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel. Basic data cleaning techniques, including trimming inconsistent text values and verifying category names, were also applied to improve uniformity. Since the dataset consists of a single structured table, no relationship modeling between multiple tables was required. The transformed dataset served as a clean, consistent, and analysis-ready foundation for developing DAX measures, KPI cards, interactive charts, slicers, and dashboards that effectively evaluate influencer performance, audience engagement, content effectiveness, and creator distribution across different categories.

.

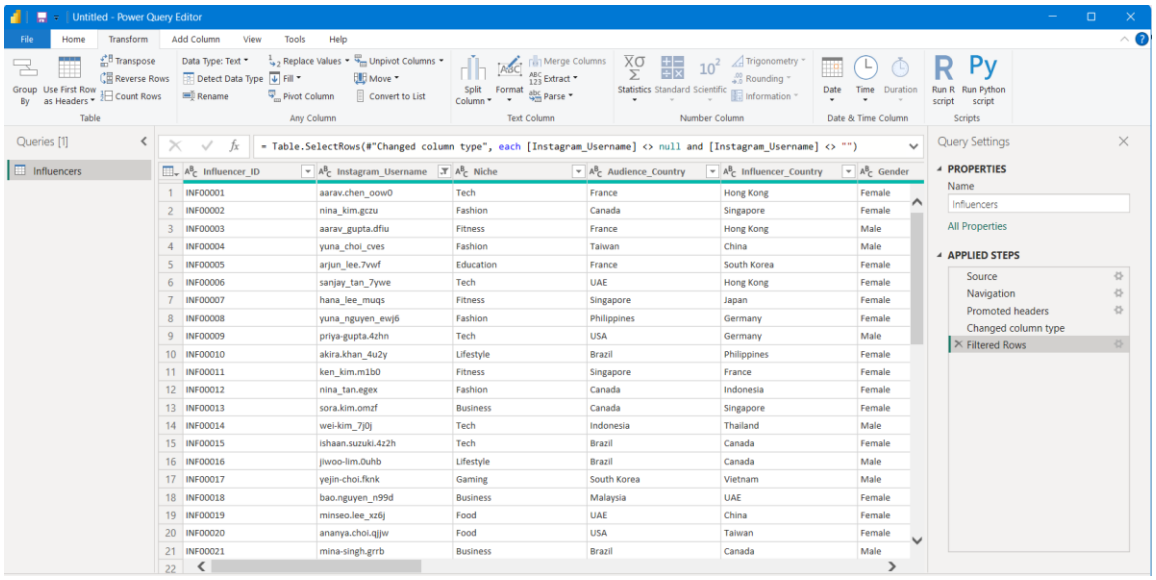


Figure 3.4.1: Power Query Editor Showing Data Transformation

Figure 3.4.1 illustrates the Power Query Editor in Microsoft Power BI used for transforming the Instagram Influencers dataset. The dataset was imported from an Excel workbook, and transformation steps such as promoting column headers, assigning appropriate data types, and filtering records with valid Instagram usernames were performed. The Applied Steps pane displays the sequence of transformations, including Source, Navigation, Promoted Headers, Changed Column Type, and Filtered Rows, ensuring that the dataset is clean, consistent, and ready for data modeling and dashboard development.

3.5 DAX Measures

Data Analysis Expressions (DAX) were used to create the KPI values and analytical calculations used throughout the Instagram Influencer Categorical Performance Analysis dashboard. All measures were defined centrally in a dedicated _Measures table to ensure consistency and accuracy across the report. These measures support the analysis of creator performance, audience quality, engagement, verification status, and content effectiveness. Centralizing calculations improves report maintainability and ensures that all visuals and KPI cards display consistent results

Table 3.3.4: Overview DAX Measures Used in the Dashboard

S.No	Measure Name	DAX Formula
1	Total Creators	COUNTROWS(Influencers)
2	Avg Followers	AVERAGE(Influencers[Followers])
3	High Quality Audience Creators	CALCULATE(COUNTROWS(Influencers), Influencers[Audience_Quality] IN {"Good","Excellent"})
4	High Quality Audience %	DIVIDE([High Quality Audience Creators],[Total Creators],0)
5	Average Engagement Rate	DIVIDE(AVERAGE(Influencers[Engagement_Rate (%)]),100)
6	Verified Creators	CALCULATE(COUNTROWS(Influencers), Influencers[Verification_Status]="Verified")
7	Avg Authentic Engagement	AVERAGE(Influencers[Authentic_Engagement])

Table 3.3.4 presents the key DAX measures created in Microsoft Power BI to support the Instagram Influencer Performance Analysis Dashboard. These measures are used to calculate essential KPIs such as the total number of creators, average followers, average engagement rate, verified creators, audience quality, and authentic engagement. The measures enable dynamic filtering, category-wise comparison, and real-time performance analysis, forming the basis for interactive dashboard visualizations and data-driven marketing insights.

4. INFLUENCER PERFORMANCE ANALYSIS

4.1 Dashboard Design

The dashboard was designed in Microsoft Power BI to provide a clear, interactive, and user-friendly interface for analyzing Instagram influencer performance. A consistent layout and visual theme were maintained across all eleven report pages, comprising one overall dashboard and ten category-specific dashboards for Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel. This consistent design enables users to navigate between pages easily while maintaining analytical continuity.

Each dashboard features a title banner at the top, followed by a row of KPI cards that provide a quick summary of key performance indicators, including Total Creators, Average Followers, High-Quality Audience Percentage, Average Engagement Rate, Verified Creators, and Average Authentic Engagement. These KPIs offer an immediate overview of influencer performance without requiring users to explore individual visualizations.

Interactive slicers are positioned on the left side of the dashboard to enable dynamic filtering of the analysis. Users can filter the report based on Age Group, Gender, Influencer Category, Creator Tier, and Verification Status, allowing customized exploration of influencer data across different audience segments.

The main dashboard area contains analytical visualizations arranged in a structured grid layout. The report includes a clustered bar chart for benchmarking engagement rates across influencer categories, a combination chart comparing followers and engagement across creator tiers, a donut chart showing gender distribution, a matrix visual analyzing content format performance by verification status, a map visual illustrating global audience reach, and a funnel chart displaying creator tier distribution. Each visualization is positioned to maximize readability while supporting interactive cross-filtering throughout the dashboard.

The overall dashboard design emphasizes simplicity, consistency, and interactivity by combining KPI cards, slicers, and analytical visuals within a well-organized layout. This design enables users to efficiently monitor influencer performance, compare engagement metrics, analyze

audience characteristics, and derive meaningful insights from the dataset through a comprehensive Business Intelligence solution.

4.2 Dashboard Creation

The dashboard was developed using a single Instagram influencer dataset containing creator demographics, audience characteristics, engagement metrics, content preferences, verification status, and audience quality information. Before importing the dataset into Microsoft Power BI, data preparation and transformation were carried out in Power Query to clean the data, remove inconsistencies, assign appropriate data types, validate records, and ensure that all fields were suitable for analysis. A dedicated Measures table was created to store DAX calculations, providing a centralized location for KPI measures and ensuring consistency, reusability, and accuracy throughout the report. Various visualizations were selected based on the analytical objectives of the project, including KPI cards, clustered bar charts, donut charts, line and column charts, matrix tables, maps, funnel charts, and interactive slicers. The report is organized into an Overview Dashboard and ten niche-specific dashboards that individually analyze Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel influencers. Each dashboard highlights key performance indicators such as creator count, follower reach, average engagement rate, authentic engagement, audience quality, and verification status to provide a comprehensive view of influencer performance. Navigation buttons were incorporated to enable seamless movement between the overview and category-specific dashboard pages, ensuring a smooth user experience. Interactive slicers and cross-filtering capabilities allow users to explore influencer data based on audience demographics, creator tier, content format, verification status, and engagement metrics. Tooltips and data labels were included to improve the clarity and interpretation of each visualization, enabling users to gain deeper insights without navigating away from the report.

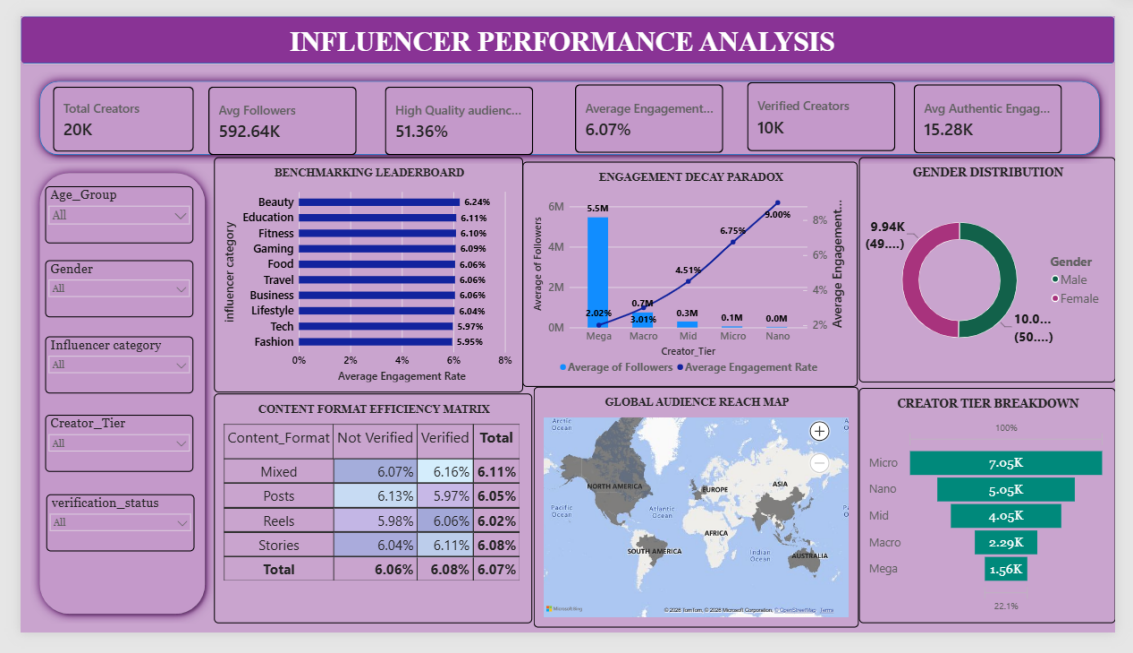


Fig 4.2.1: Instagram Influencer Performance Analysis

Figure 4.2.1 presents the Instagram Influencer Categorical Performance Analysis Dashboard developed in Microsoft Power BI. The dashboard provides a comprehensive analysis of influencer performance across multiple categories by displaying creator metrics, audience quality indicators, engagement statistics, verification insights, content format effectiveness, and creator tier distribution. Interactive slicers enable users to filter and explore data based on demographic, behavioral, and performance attributes, supporting detailed and data-driven influencer analysis.

KPI Cards

The Influencer Performance Analysis page provides a comprehensive overview of influencer performance by integrating key performance indicators (KPIs)

- Total Creators → Displays the total number of Instagram creators.
- Average Followers → Shows the average follower count of creators.
- High Quality Audience % → Indicates the percentage of creators with high-quality audiences.
- Average Engagement Rate → Represents the average engagement rate across creators.

- ❖ Verified Creators → Displays the total number of verified creators.
- ❖ Average Authentic Engagement → Shows the average authentic engagement of creators.

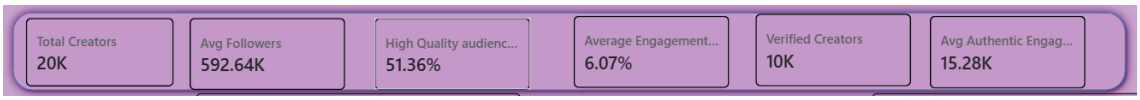


Figure 4.2.2 KPI Cards – Overall Influencer Performance

Figure 4.2.2 presents the key performance indicators that summarize the overall influencer dataset. The KPI section displays Total Creators, Average Followers, High Quality Audience Percentage, Average Engagement Rate, Verified Creators, and Average Authentic Engagement. These metrics provide users with an instant overview of creator performance, audience quality, and engagement statistics before performing detailed analysis.

Benchmarking Leaderboard

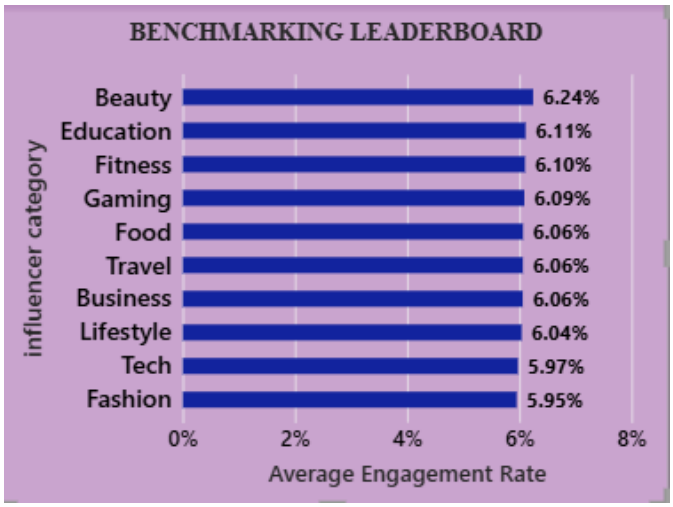


Fig. 4.2.3: Benchmarking Leaderboard

Figure 4.2.3 compares the average engagement rate across different influencer categories. The visualization highlights the relative performance of each category, allowing users to identify categories with higher audience engagement. This comparison supports benchmarking and evaluation of influencer effectiveness across multiple niches.

Engagement Decay Paradox

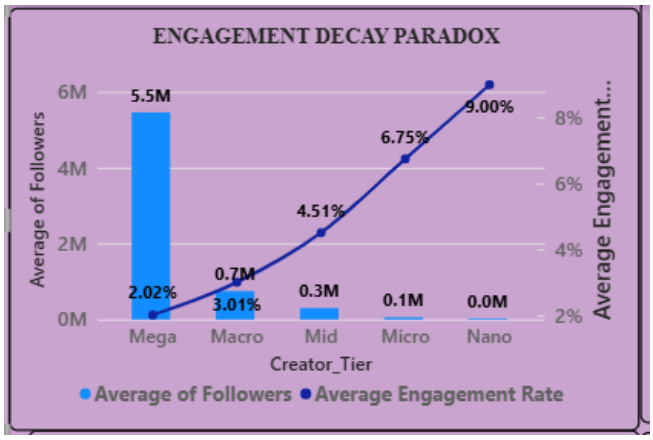


Fig. 4.2.4: Engagement Decay Paradox

Figure 4.2.4 presents the relationship between creator tiers, average followers, and average engagement rate across Instagram influencers. It illustrates how follower count changes from Nano to Mega creators while simultaneously comparing their corresponding engagement rates. The visualization helps identify engagement patterns across different creator tiers and demonstrates that creators with larger audiences do not necessarily achieve higher engagement levels, supporting more effective influencer performance evaluation.

Gender Distribution

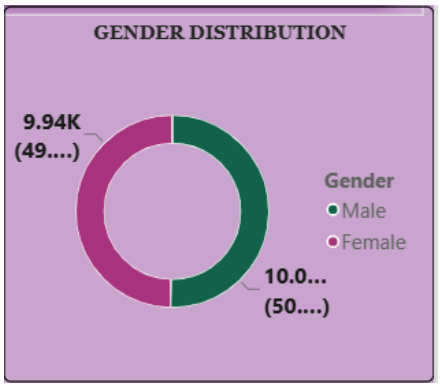


Fig. 4.2.5: Gender Distribution

Figure 4.2.5 presents the Gender Distribution illustrating the proportion of influencers based on gender categories. Each segment of the chart represents a specific gender group, enabling

users to understand the overall demographic composition of creators within the dataset and compare the representation of different gender groups.

Content Format Efficiency Matrix

CONTENT FORMAT EFFICIENCY MATRIX

Content_Format	Not Verified	Verified	Total
Mixed	6.07%	6.16%	**6.11%**
Posts	6.13%	5.97%	**6.05%**
Reels	5.98%	6.06%	**6.02%**
Stories	6.04%	6.11%	**6.08%**
Total	**6.06%**	**6.08%**	**6.07%**

Fig. 4.2.6: Content Format Efficiency Matrix

Figure 4.2.6 presents the Content Format Efficiency Matrix, which evaluates engagement performance across different content formats and verification statuses. The visual compares engagement rates for Mixed, Posts, Reels, and Stories while distinguishing between verified and non-verified creators. This analysis helps identify the most effective content strategies for maximizing audience engagement.

Global Audience Reach Map



Fig. 4.2.7: Global Audience Reach Map

Figure 4.2.7 presents the Global Audience Reach Map, which displays the geographical distribution of influencer audiences across different countries. The visualization enables users to identify regions with significant audience presence and understand the global reach of influencer content.

Creator Tier Breakdown

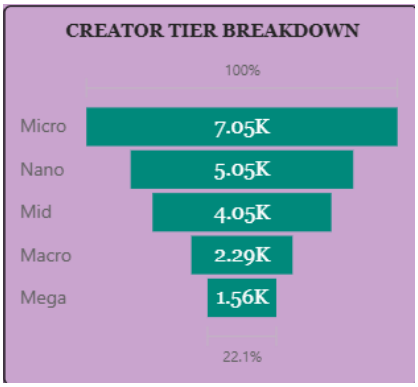


Fig. 4.2.8: Creator Tier Breakdown

Figure 4.2.8 presents the Creator Tier Breakdown, illustrating the distribution of influencers across Mega, Macro, Mid, Nano, and Micro creator tiers. The visualization enables users to compare the size of each creator segment and understand the overall distribution of creators within the dataset.

5. RESULTS & DISCUSSION

The Instagram Influencer Categorical Performance Analysis dashboard successfully transformed influencer data into meaningful insights through interactive visualizations and KPI-based analysis. The dashboard revealed significant variations in engagement performance, audience quality, creator reach, and content effectiveness across different influencer categories.

The analysis identified Beauty as the highest-performing category in terms of average engagement rate, while Fashion recorded comparatively lower engagement levels despite having a large creator base. The findings also showed that audience quality varies across categories, indicating that follower count alone is not a reliable measure of influencer effectiveness.

Creator tier analysis demonstrated that smaller creators, particularly Nano and Micro influencers, often achieve higher engagement rates than larger creator tiers, highlighting the importance of audience connection and authenticity. The dashboard further revealed differences in content format performance, with certain formats generating stronger audience interaction and authentic engagement than others.

Demographic and audience analysis provided insights into gender distribution, age-group performance, audience reach, and verification patterns. The global audience analysis showed that influencer content reaches diverse geographical markets, emphasizing the international nature of influencer marketing.

Overall, the dashboard enabled efficient comparison of influencer categories and performance metrics, providing a data-driven foundation for evaluating influencer effectiveness and supporting strategic marketing decisions.

6. CONCLUSION

Instagram Influencer Categorical Performance Analysis Dashboard was successfully developed using Microsoft Power BI to transform influencer profile, audience, and engagement data into an interactive Business Intelligence solution. Using a dataset of 20,000 Instagram influencer records, the dashboard integrates data preparation, Power Query transformations, DAX measures, and interactive visualizations to provide meaningful insights into influencer performance across ten categories: Beauty, Business, Education, Fashion, Fitness, Food, Gaming, Lifestyle, Technology, and Travel. The dashboard enables users to evaluate influencer performance, compare audience engagement patterns, and identify category-specific trends through dynamic filtering and interactive reports.

The completed dashboard eliminates the need for manual spreadsheet-based analysis by providing a centralized reporting environment where users can explore creator reach, engagement rate, authentic engagement, audience quality, creator tiers, content formats, verification status, and audience demographics. The Executive Performance Analysis page provides an overall summary of influencer performance, while the category-specific dashboards enable detailed analysis of individual influencer categories.

The analysis demonstrates that influencer performance depends on multiple factors beyond follower count, including engagement rate, audience quality, content format, creator tier, and demographic characteristics. The insights generated by the dashboard help identify high-performing categories, effective content strategies, and audience engagement patterns, supporting informed influencer marketing decisions. The dashboard successfully achieves its objective of providing a comprehensive platform for influencer performance analysis and supports data-driven decision-making in influencer marketing and digital brand strategy.

7. REFERENCES

1. Freberg, K. et.al [5] (2011), "Who are the social media influencers? A study of public perceptions of personality," *Public Relations Review*, vol. 37, no. 1, pp. 90–92, 2011.
2. S. Yuan, C. et.al [6] "Influencer marketing: How message value and credibility affect consumer trust of branded content on social media," *Journal of Interactive Advertising*, vol. 19, no. 1, pp. 58–73, 2019.
3. M. De Veirman, et.al [7] "Marketing through Instagram influencers: The impact of number of followers and product divergence on brand attitude," *International Journal of Advertising*, vol. 36, no. 5, pp. 798–828, 2017.
4. W. Eckerson, et.al [8] *Performance Dashboards: Measuring, Monitoring, and Managing Your Business*, 2nd ed. Hoboken, NJ, USA: John Wiley & Sons, 2011.
5. M. Russo, et.al [9], *The Definitive Guide to DAX: Business Intelligence for Microsoft Power BI, SQL Server Analysis Services, and Excel*, Redmond, WA, USA: Microsoft Press, 2015.
6. Microsoft Corporation. (2026). *Power BI Documentation*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/power-bi/
7. Microsoft Corporation. (2026). *Data Analysis Expressions (DAX) Reference*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/dax/
8. Microsoft Corporation. (2026). *Power Query Documentation*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/power-query/
9. Microsoft Corporation. (2026). *Power BI Desktop Documentation*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/power-bi/fundamentals/